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Constructing Set-Compositional and Negated Representations for First-Stage Ranking

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Abstract

Set compositional and negated queries are crucial for expressing complex information needs and enable the discovery of niche items like “Books about non-European monarchs”. Despite the recent advances in LLMs, first-stage ranking remains challenging due to the requirement of encoding documents and queries independently from each other. This limitation calls for constructing compositional query representations that encapsulate logical operations or negations, and can be used to match relevant documents effectively. In the first part of this work, we explore constructing such representations in a zero-shot setting using vector operations between lexically grounded Learned Sparse Retrieval (LSR) representations. Specifically, we introduce *Disentangled Negation* that penalizes only the negated parts of a query, and a *Combined Pseudo-Term* approach that enhances LSR’s ability to handle intersections. We find that our zero-shot approach is competitive and often outperforms retrievers fine-tuned on compositional data, highlighting certain limitations of LSR and Dense Retrievers. Finally, we address some of these limitations and improve LSR’s representation power for negation, by allowing them to attribute negative term scores and effectively penalize documents containing the negated terms.¹

CCS Concepts

• **Information systems** → **Information retrieval; Retrieval models and ranking; Information retrieval query processing.**

Keywords

information retrieval, learned sparse retrieval, negations, intersections, set-compositionality

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¹We open source our code at <https://github.com/littlewine/compositional-LSR>



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1 Introduction

Information retrieval (IR) systems have become increasingly important in the era of Large Language Models (LLMs) [2, 20, 28]. Search plays a crucial role in helping LLMs answer user questions or engage in information-seeking conversations by providing evidence to ground or verify their responses, thereby reducing the risk of hallucinations. As these technologies and applications evolve, search queries become increasingly complex [51]. A common type of complexity occurs when a query includes multiple constraints, often combined using logical operations like conjunctions (“and”), disjunctions (“or”), or negations (“not”). For example, a user might search for books about French monarchs excluding Napoleon, or a documentary about Canada filmed in Mississippi. These complex queries help users discover niche items, and can support tasks like known-item retrieval [1, 5, 16] or conversational search and clarifying questions [26].

Traditionally, IR systems used Boolean queries with logical operations [14], but those were superseded by probabilistic and vector space models without any Boolean logic. Modern SOTA Neural IR retrievers also lack Boolean logic since they are pre-trained on large-scale ad-hoc training datasets such as MSMarco [36] that contain few such logical operators and negations, and little work has been done to evaluate their effectiveness there.

Recent research found that even SOTA retrievers struggle with conjunctions and negations [31, 47]. For negations, prior work has shown that a major challenge is the interference between the positive and negative parts of a query, often leading negation methods to reduce the ranking scores of relevant documents [10, 48]. First-stage ranking is especially challenging because queries and documents must be encoded independently (i.e., without cross-attention) into vectors before matching via a similarity metric. This limitation necessitates that vector representations must effectively encapsulate this compositionality. In Dense Retrieval (DR), most work has focused on negation; Widdows [48] introduced Orthogonal Negation, a method aimed at removing only the negated components of a dense vector. For lexical retrieval, most methods focused on applying Rocchio relevance feedback [40] to “dampen” the appearance of negated terms in retrieved documents [44, 53].

This work focuses on lexical representations of queries and documents. Lexical methods are interpretable because they build representations that are grounded in a vocabulary. This flexibility enables the design of intuitive operations on the representations that encapsulate set-compositionality. Meanwhile, recent research has overcome most limitations of traditional lexical methods, i.e. the vocabulary mismatch and the ability to do supervised learning [13, 29, 32, 35, 54], with Learned Sparse Retrievers (LSR) estimating term importance in the context of a specific query or document.

As a consequence, LSR models are very effective first-stage rankers (SOTA in out-of-domain settings and competitive to DR in-domain), while preserving this lexical grounding in their representations.

We first propose a zero-shot framework that constructs set-compositional representations of queries by performing Linear Algebra Operations (LAO) between LSR representations. We design LAO that manipulate sparse representations to reflect logical operations. For negations, we propose the method of *Disentangled Negation*, that takes advantage of this lexical grounding and the "disentanglement" between different dimensions of the representation to mitigate the pressing problem of interference between positive and negative aspects of a query [10, 48]. Regarding intersections, we identify the shortcoming of current LSR representations in capturing joint term occurrences and conclude this has a positive impact in ranking. Such zero-shot methods are useful because they work with existing pre-trained models without additional finetuning, offering finer control over ranking or aiding in collecting training data (e.g., top-K pooling [46]) to train compositional retrievers.

Beyond zero-shot methods, we examine whether retrievers can learn such compositional representations through training on the ranking task. Both LSR and Dense retrievers perform well on negations but not on intersections. Their performance is comparable to zero-shot LAO, while Disentangled Negation shows a slight advantage at top ranks. We hypothesize this stems from a limitation of supervised LSR models: they can remove negated terms from queries but cannot penalize documents containing them.

To address this limitation, we adapt the architecture of LSR models to allow them attribute negative term scores. Our experiments demonstrate that in settings where enough training data exists, this is very effective and leads to large improvements compared to traditional LSR models that do not produce negative weights. For negations, we observe an advantage of methods that have the capacity to score negatively (ie. penalize) terms. Intrigued by this observation, we explore to what extent different compositional types can be learned from training data for different retrieval families. We conclude that both Dense (DR) and Learned Sparse Retrievers (LSR) do benefit from training on negations, but somewhat surprisingly cannot learn how to handle Intersections (at least under domain shift).

In particular, we make the following contributions:

- We propose a zero-shot framework that constructs set - compositional representations of queries by performing Linear Algebra Operations (LAO) between atomic representations. Most importantly, we introduce the LAO of *Disentangled Negation*, that penalizes negated terms of a query without affecting the positive ones, and Combined Pseudo-Terms, that expands the representation capabilities of LSR models when modelling Intersections.
- We explore the ability of Dense and Learned Sparse Retrievers to learn compositional representations, concluding that this is effective for Set Difference, but none of the models is able to do so for Intersections.
- We improve the representation power of Learned Sparse Retrievers (LSR) for negated queries, by allowing them to attribute negative term scores and penalize documents containing the negated terms.

To the best of our knowledge, this is the first work that explores constructing set-compositional and negated representations using LSR models that are lexically grounded, SOTA first-stage rankers.

2 Related Work

We first discuss the use of logical operators in traditional IR systems, followed by related work on compositionality and transformers, and close this section with works discussing negations for IR.

Logical operations in IR. Early Information Retrieval systems used Boolean search, employing logical operators to construct queries and retrieve documents based on term disjunction, conjunction, or absence [33, 42, 43]. However, these systems didn't rank results, returning only a document set, and required a complex query language to express queries [14, 42, 43], ultimately declining in popularity as IR shifted to statistical methods like tf-idf and BM25 [17, 39].

Later approaches attempted to incorporate compositionality and logic operations by performing linear algebra on dense vector representations [48, 49] or applying relevance feedback to lexical methods [4, 44, 50, 53]. Among the former, Widdows [48] introduced orthogonal negation, which subtracts the projection of a negative vector from the positive vector. Experiments showed that this method is effective in removing negated concepts, but may still negatively impact retrieval of the positive ones. Widdows et al. [49] surveyed various vector composition methods, distinguishing between explicit approaches (e.g., orthogonal negation) and implicit methods (e.g., attention in neural networks). Regarding relevance feedback methods, Salton and Buckley [44] used Rocchio feedback [40] to penalize terms found in non-relevant document sets and handle negations. Zhang et al. [53] proposed a similar method for session search, introducing an algorithm that models a session using a theme query vector $\vec{q}$, along with positive ($+\Delta\vec{q}$) and negative ($-\Delta\vec{q}$) query changes based on user document clicks and query modifications. To the best of our knowledge, very few studies investigate compositionality and negations for modern neural IR models. We draw inspiration from these works and explore how some of their ideas can be applied to LSR, that are state-of-the-art first-stage rankers.

Compositionality and negation in Transformers. Modern Neural IR models are based on the transformer architecture [13, 19, 21, 23, 45]. However, transformers' ability to tackle compositionality and negations has been widely debated in the NLP community, with many works highlighting their struggle in doing so [3, 11, 15, 18, 22, 37]. Our work differs in two key axes. First, we explore set-compositionality specifically for ranking, rather than tasks involving numerical operations or problem decomposition. Second, we focus on first-stage ranking, where documents and queries must be encoded independently. This constraint requires models to encapsulate compositionality within a single vector, without relying on multiple layers of cross-attention. Word2Box [7] introduced n-dimensional hyperrectangle (box) embeddings, showing that region-based approaches outperform traditional embeddings for set-compositionality and polysemy in word similarity benchmarks. However, these would require integration over boxes to compute query and document scores, which makes them difficult to apply to first-stage ranking that relies on kNN search [9, 52].

Therefore, we focus on the first-stage setting and investigate the widely used DR and LSR models in the context of Neural IR.

Challenges for negations in IR. Previous work discusses the effectiveness of negation in IR and highlights challenges therein. They showed that applying negations can reduce method effectiveness, as negating a concept closely linked to the target information need significantly affects relevant items ranking scores [10, 48]. To address this, a set of methods try to limit the impact of negated terms or aspects in a query, controlling it with a hyperparameter (Constant subtraction) [24, 25, 48]. Research in conversational and clinical IR shows that, contrary to intuition, including negated terms can positively impact retrieval [24, 26].

Lastly, McQuire and Eastman [34] discusses the problem of ambiguity in queries with negation. They show that when such queries increase in complexity, disagreements arise on attributing the negation between human annotators, and automatic systems failed as well. In this work, we build on these findings and further investigate negations in the context of Neural IR and LSR models.

3 Methodology

In this section we outline our methodology. We start with preliminaries on LSR (Section 3.1), followed by a framework for deriving set-compositional representations (Section 3.2). On Section 3.3 we introduce an adaptation of LSR for modelling negations.

3.1 Learned Sparse Retrievers

Learned Sparse Retrieval (LSR) is a family of efficient and effective Neural IR models. They improve on traditional lexical methods (eg, BM25 [39]) by (a) learning an expansion over the input text, closing the vocabulary mismatch, and (b) predicting term importance scores. Splade [13] does so by computing term importance scores w_j for a token t_j at position j over a WordPiece vocabulary. To get w_j , input embeddings are passed through multiple transformer layers, deriving a token representation h_i at input position i . Finally, an MLM head (E, b) computes a weight out_{ij} corresponding to the importance of vocabulary term j given input term i .

$$out_{ij} = \text{transform}(h_i)^T \cdot E_j + b_j \quad j \in \{1, \dots, |V|\} \quad (1)$$

A Rectified Linear Unit and a logarithmic function ensure term scores are positive with a log-saturation effect. Finally, scores are aggregated over all input tokens with a Max Pooling operation:

$$w_j = \max_{i \in t} \log(1 + \text{ReLU}(out_{ij})) \quad (2)$$

In practice, this happens in a query or document encoder f_{QE}, f_{DE} that maps the input text to a sparse representation $\vec{Q}$ or $\vec{D}$.

$$\begin{aligned} \vec{Q} &= f_{QE}(Q) = [w_0^q, w_1^q, \dots, w_{|V|}^q] \\ \vec{D} &= f_{DE}(D) = [w_0^d, w_1^d, \dots, w_{|V|}^d] \end{aligned} \quad (3)$$

For brevity, these representations can also be written as $\vec{Q} = (t_j : w_j, \dots) \quad \forall w_j \neq 0$. The ranking score between a query and document is computed by multiplying the term scores between Q and D across vocabulary V .

$$\text{score}(Q, D) = \sum_{i=1}^{|V|} w_i^q \cdot w_i^d \quad (4)$$

3.2 Deriving compositional query representations with Linear Algebra Operations (LAO)

In this part we describe a framework (Figure 1) to derive representations of set-compositional queries, based on their atomic queries and set operations (Section 3.2.1). The framework relies on Linear Algebra Operations (LAO) that are designed to tackle a particular logical operations (Section 3.2.2).

Notation. Let us consider an atomic query q_A , with a sparse query representation vector $\vec{A} = [w_0^A, w_1^A, \dots, w_{|V|}^A]$ and let $|A|$ be the set of relevant documents to q_A . Let $q_{A \star B}$ be a compositional query, where $\star \in \{\setminus, \cup, \cap, \}$ for which we seek to construct a representation $\vec{Q}_{A \star B}$ that is effective in matching the relevant documents $|A \star B|$. For instance, consider the atomic queries $q_A = \text{"BIRDS OF COLOMBIA"}$ and $q_B = \text{"BIRDS OF VENEZUELA"}$ with the binary query representation vectors $\vec{A} = (\text{BIRDS:1, FLY:1, COLOMBIA:1, ANDES:1})$ and $\vec{B} = (\text{BIRDS:1, FLY:1, VENEZUELA:1, ANDES:1})$.

3.2.1 A framework for deriving zero-shot compositional representations. Current query encoders have been pre-trained on large ad-hoc query datasets that contain few set-compositional queries with *union*, *intersection* and *set difference*. Therefore, our hypothesis is that they can construct effective representations for atomic queries, which we can later combine using LAO designed to reflect the logical set-operations (further motivated in Section 3.2.2).

Our framework requires an explicit formulation of set-compositional queries, ie. the atomic queries and logical operations. For the query $q_{A \cap B} = (\text{BIRDS OF COLOMBIA}) \text{ THAT ARE ALSO } (\text{BIRDS OF VENEZUELA})$, a decomposition is required to derive atomic queries $A = (\text{BIRDS OF COLOMBIA})$, $B = (\text{BIRDS OF VENEZUELA})$, and the set-compositional operation of *Intersection* ($A \cap B$). We do not investigate how to derive this decomposition here, as preliminary experiments showed that in this dataset, LLMs can perform this task very accurately. Instead, we use the atomic queries and operations provided directly by the QUEST dataset. Then, we use an LSR query encoder f_{QE} to construct the atomic query representations $\vec{A}, \vec{B}$ from the corresponding atomic queries q_A, q_B :

$$\vec{A} = f_{QE}(q_A)$$

Then, we use a compositional query encoder f_{CQE} to combine the atomic embeddings using linear algebra operations that reflect the logical set operations.

$$f_{CQE}(q_A \star q_B) = \vec{A} \circ \vec{B} \quad , \text{ where } \star \in \{\cap, \cup, \setminus\}$$

and $\circ$ is the corresponding LAO described in Subsection 3.2.2.

3.2.2 Linear Algebra Operations (LAO) for set-theoretic operations. In this part, we define LAO between sparse embeddings, that reflect the set operations of *union*, *intersection* and *difference*.

Set Difference ($q_A \setminus q_B$). Set difference is closely related with negation, since it includes a positive and a negative part within a query. Based on set theory, we are looking for all elements (documents) of A that are not members of B , ie. $|A \setminus B| = |A| - |A \cap B|$. An intuitive formulation to representing set difference would be *Subtraction*, penalizing documents containing terms related to q_B and therefore $\vec{B}$:

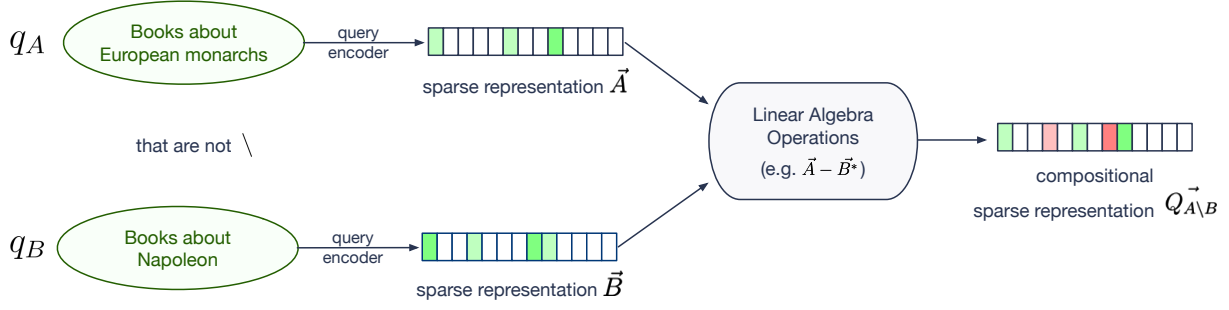


Figure 1: A zero-shot framework for constructing compositional query representations with Linear Algebra Operations (RQ1).

$$f_{CQE}(A \setminus B) = \vec{A} - \vec{B}$$

Considering the previous example, simple Subtraction would result in a representation of (COLOMBIA:1, VENEZUELA:-1), penalizing documents mentioning VENEZUELA, but also removing BIRDS and FLY. Prior work highlighted the difficulty of building negated representations using subtraction, due to an interference between positive and negative query [10, 48]. Intuitively, Widdows [48] explains that when queries q_A, q_B have high semantic overlap, negating $\vec{B}$ inevitably pushes A away from its original representation. LSR representations present an opportunity to tackle this problem, since they are sparse and lexically grounded, as each dimension i preserves a semantic relationship to token t^i . Therefore LSR representations are much more disentangled compared to DR, allowing us to formulate a representation that only penalizes dimensions present in B without affecting dimensions relevant to A , expressing *Disentangled Negation*:

$$f_{CQE}(A \setminus B) = \vec{A} - \vec{B}^*, \text{ where } \vec{B}^* = \vec{B} - \mathbb{1}_A \odot \vec{B} \quad (5)$$

The final representation (COLOMBIA:1, VENEZUELA:-1, BIRDS:1, FLY:1, ANDES:1), manages to penalize terms related to negative query aspects while retaining all of the positive ones.

Intersection ($q_A \cap q_B$). Expressing query term intersections in LSR can be challenging, due to the semantic relationship of each dimension to a token. Our hypothesis is that for modeling intersections, it would be useful to expand the representation vector from $|V|$ to $|V|^2$ and create **Combined Pseudo-Terms (CPT)**, capturing joint term-occurrence such as COLOMBIA $\cap$ VENEZUELA. In practice, this is an outer product between query vectors $\vec{A}$ and $\vec{B}$:

$$f_{CQE}(A \cap B) = \vec{A} \otimes \vec{B} = \begin{bmatrix} w_{A^0 B^0} & w_{A^0 B^1} & \dots & w_{A^0 B^{|V|}} \\ w_{A^1 B^0} & w_{A^1 B^1} & \dots & w_{A^1 B^{|V|}} \\ \dots & \dots & \dots & \dots \\ w_{A^{|V|} B^0} & \dots & \dots & w_{A^{|V|} B^{|V|}} \end{bmatrix} \quad (6)$$

where $w_{A^i B^j} = \sqrt{w_A^i \cdot w_B^j}$. For efficiency reasons, we limit the number of dimensions (top-terms per compositional query) to 5.

Similarly, we represent documents with the outer product of the document vector:

$$f_{CDE}(D) = \vec{D} \otimes \vec{D}, \text{ where } w_{D^i D^j} = \sqrt{w_D^i \cdot w_D^j} \quad (7)$$

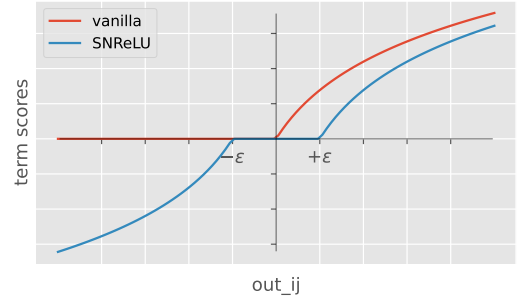


Figure 2: Splade activation functions.

Finally, we compute the *Combined Pseudo-Scores* of queries and documents as:

$$CPTscore(A \cap B, D) = ||f_{CQE}(A \cap B) \cdot f_{CDE}(D)|| \quad (8)$$

The quadratic expansion of dimensions expands the index size, but can be supported by modern inverted indexes designed for LSR models [6]. Alternatively, a two-step approximation selecting a collection subset (similar to [23]) and computing the expanded representation ($\vec{D} \otimes \vec{D}$) on the fly would also be possible.

Union ($q_A \cup q_B$). Based on the Inclusion-Exclusion principle [38] of set theory, ie. $|A \cup B| = |A| + |B| - |A \cap B|$, an approximate representation of union between embeddings can be formulated as *Addition*:

$$f_{CQE}(A \cup B) = \vec{A} + \vec{B} \quad (9)$$

However, set theory is concerned with Boolean set membership, while for ranking term weights w_A^i are important for calculating relevance scores. Therefore, such an approximation would tend to inflate scores of common terms between A and B , as those would be counted twice. To avoid this, we also consider a *MaxPooling* operation for union between atomic embeddings:

$$f_{CQE}(A \cup B) = \text{MaxPool}(\vec{A}, \vec{B}) \quad (10)$$

3.3 Improving LSR representations for negations

Traditionally, lexical and LSR models use positive term weights (w_i) to compute ranking scores between queries and documents (Eq. 4). We hypothesize that this limits their ability to handle negations, as they can't represent and penalize negated terms in documents.

To address this, we modify the architecture of Splade, a SOTA LSR model, to allow **negative** term weights for representing negated aspects in queries or documents. We design a new activation function with properties similar to Splade’s (Eq. 2), such as the log saturation effect and sparsity, while also supporting negative term weights. Specifically, we introduce SNReLU, a symmetrical-odd version of Splade’s activation function, which enables term sparsity (0 term scores) by shifting the original function by $-\epsilon$ and $+\epsilon$.

$$\begin{aligned} out_pos_{ij} &= \log(1 + \text{ReLU}(out_{ij} - \epsilon)) \\ out_neg_{ij} &= -\log(1 + \text{ReLU}(-out_{ij} + \epsilon)) \end{aligned} \quad (11)$$

We propose to aggregate token scores over the input in two different ways. The first selects the input position with the largest absolute value ($\pm|max|$, Eq. 12). Alternatively, we first pool the maximum positive and minimum negative values for each token, and then summing the two (sum Eq. 13).

$$w_j = \begin{cases} \max_{i \in t}(out_pos_{ij}) & , \text{ if } \max_{i \in t}|out_pos_{ij}| \geq \max_{i \in t}|out_neg_{ij}| \\ \min_{i \in t}(out_min_{ij}) & , \text{ otherwise} \end{cases} \quad (12)$$

$$w_j = \max_{i \in t}(out_pos_{ij}) + \min_{i \in t}(out_min_{ij}) \quad (13)$$

4 Experimental Setup

In this section we describe our experimental setup.

4.1 Datasets and Evaluation

QUEST. QUEST [31] is an entity retrieval dataset containing set-compositional queries (eg. “Ming dynasty or Chinese historical novels”). The dataset is build from multiple atomic queries, ie. individual atomic constraints that are combined using *set difference*, *union*, or *intersection*. Each query contains a template version that verbalizes the set operation using templates (eg. “X that are also Y” for intersection) and a human-rephrased version. Entity categories (eg. “Ming dynasty novels”) are used as atomic queries and composed into a compositional template-based query, using templates. Human annotators paraphrase the template-based queries, while also assessing their fluency and naturalness. Throughout this work, we use the template versions of the queries, as during preliminary experiments we found no significant difference between the two. We evaluate performance using $nDCG@10$, that evaluates retrieval effectiveness while giving more importance to the top ranking positions, and $R@100$, that evaluates what percentage of relevant documents is retrieved within the top-100 documents.

Train-test split across domains. During initial experiments, we identified a leakage between the original train and test splits. Although compositional queries were unique across sets, their atomic queries appeared in both, and even when atomic queries were unique, they often had substantial overlap with other atomic queries. For example, train query “Books about monarchs but not about France” is very closely related to the test query “Books about monarchs that don’t include Napoleon”, creating an overlap of 75% between the two relevance sets. In such cases, it is not clear whether

the retriever learns to identify relevance and construct set compositional representations, or simply memorizes associations between atomic queries and relevant documents. To solve this issue we perform a domain split ensuring each entity category appears either in train or test set, keeping all “Film and “Book” category queries in the training set and “Plants” and “Animals” in the test set.

NevIR. NevIR [47] is a benchmark aiming to evaluate how negation impacts Neural Information Retrieval. It contains synthetic counterfactual query and document pairs, build on top of minimally contrasting documents. Contrasting documents are identical with only a clause being different, for instance: “... GENERALLY, DRUG POSSESSION IS AN ARRESTABLE OFFENSE, ...” vs. “...HOWEVER, RARELY IS DRUG POSSESSION AN ARRESTABLE OFFENSE...”. Then, two queries are constructed to be relevant to one of the two documents. Following the evaluation setup of Weller et al. [47], we use *pair-wise accuracy* for evaluation, that measures whether both counterfactual queries scored the relevant document higher than the irrelevant.

4.2 Retrieval models and Baselines

First we describe the retrieval models used in our experiments, followed by baseline and ablation methods.

4.2.1 Retrieval Models. We include representative models from different families of retrieval models, specifically:

- BM25 [39], a SOTA lexical retriever (using pyserini [30])
- Splade [12, 13, 27], a SOTA Learned Sparse Retriever
- Contriever [19], a SOTA Dense Retriever

We start from the splade-v2² and contriever³ checkpoints, unless stated otherwise. We perform a domain pre-finetuning step on the non-compositional subset of QUEST (questNC), to adapt models from the distribution of MSMarco to QUEST (eg. Wikipedia documents instead of marco passage collection). Note that during this step we only use atomic queries and not compositional queries.

4.2.2 Compositionality-aware baselines and ablations. We use appropriate baselines and linear algebra operation ablations designed for each logical set-operation.

Baselines.

- *vanilla*: Verbalizes the set operation using a template per set operation (eg. “ q_A that are not q_B ”)
- *NRF* [44]: Negative relevance feedback is based on the Rocchio feedback algorithm [41] and uses positive and negative feedback to change the query vector away from irrelevant documents and closer to relevant. We adapt the method, using the negative query vector $\vec{B}$ as the negative feedback vector, ie. $\vec{A} - \lambda \cdot \vec{B}$.
- *Orthogonal negation* [48]: a method that subtracts the projection π of query B from query A, which in practice alters the positive vector A to remove the common aspects found in B, ie. $A \setminus B = \vec{A} - \pi_{\vec{B}} \vec{A} = \vec{A} - \frac{\vec{A} \cdot \vec{B}}{|\vec{B}|^2} \cdot \vec{B}$
- *score-fusion*: we rank documents for queries A and B independently, and fuse the query-document scores using the

²<https://huggingface.co/naver/splade-cocondenser-ensembledistil>

³<https://huggingface.co/facebook/contriever-msmarco>

corresponding operand $\circ$:

$$\text{score}(d, q_A, q_B) = \text{score}(d, q_A) \circ \text{score}(d, q_B)$$

where $\circ$ corresponds to $+$ for union, $\cdot$ for intersection and $-$ for negation. This is inspired by probability theory and based on the assumption that retrieval scores try to approximate the probability of relevance (eg. $P(A \cap B) = P(A) \cdot P(B)$)

- *score-fusion-scale*: similar to the previous, but with document scores scaled per atomic query.

Among other possible baselines, Zhang et al. [53] uses document clicks and query changes, which are unavailable in our setting. Although an adaptation to query vectors would be possible, it can be viewed as a special case of *NRF*.

LAO ablations:

- *Addition*: $\vec{A} + \vec{B}$, used for intersection and union (as motivated on Section 3.2.2)
- *Max Pool*: $\text{MaxPool}(\vec{A} + \vec{B})$, used for intersection and union (as motivated on Section 3.2.2)
- *Ignore Negation*: $\vec{A}$, used for set difference. Simply ignores the negated part of a query.

5 Results

In this Section, we discuss our experimental results and seek answers to the following Research Questions:

- RQ1** How can we build effective sparse representations of set-compositional queries in a zero-shot setting?
- RQ2** To what extent can the different families of Dense and Learned Sparse Retrievers learn set-compositional query representations?
- RQ3** How can we improve the representation power of Learned Sparse Retrievers (LSR) to account for negations?

5.1 Zero-shot compositional query retrieval with Linear Algebra Operations

In this section we try to answer **RQ1**, that aims to build zero-shot compositional representations using atomic queries and their logical operators. We apply a framework that combines atomic LSR embeddings using Linear Algebra Operations (Section 3.2).

Table 1 reports the ranking performance on QUEST for several zero-shot methods designed to handle set-compositionality. Those include the *vanilla* baseline, score-fusion methods that fuse document scores from queries q_A and q_B , and the compositional representations derived using the framework and Linear Algebra Operations (LAO) we introduced in Section 3.2. Our first observation is that pre-trained LSR models (*vanilla*) work well for Union queries, but can be improved for Set Difference and Intersection queries. Score-fusion methods are effective only on Set Difference queries, but do not outperform the simple method of ignoring the negation. From the last group of methods, we conclude that constructing compositional representations with the LAO can help for the more challenging operations of Set Difference and Intersection.

Set Difference ($A \setminus B$). For Set Difference, subtracting the LSR vectors is ineffective, but ignoring the negation works particularly well, especially in $R@100$. We hypothesize performance of this method is exaggerated due to (a) the small document collection ($\sim 300K$

method	$Q_{A \star B}$	nDCG@10	R@100
Set Difference ($q_A \setminus q_B$)			
vanilla	$f_{QE}(q_A \text{ but not } q_B)$	0.132	0.307
score-fusion (-)	-	0.228	0.332
score-fusion-scale(-)	-	0.228	0.332
Subtraction	$\vec{A} - \vec{B}$	0.120	0.190
Ignore negation	$\vec{A}$	0.233	0.390
Orthogonal negation [48]	$\vec{A} - \pi_{\vec{B}} \vec{A}$	0.235	0.379
NRF [44]	$\vec{A} - \lambda \cdot \vec{B}$	0.256	0.379
Disentangled negation (ours)	$\vec{A} - \vec{B}^*$	0.258	0.398
Union ($q_A \cup q_B$)			
vanilla	$f_{QE}(q_A \text{ or } q_B)$	0.243	0.383
score-fusion (+)	-	0.204	0.335
score-fusion-scale(+)	-	0.197	0.326
Addition	$\vec{A} + \vec{B}$	0.237	0.386
Max Pool	$\text{MaxPool}(\vec{A}, \vec{B})$	0.254	0.387
Intersection ($q_A \cap q_B$)			
vanilla	$f_{QE}(q_A \text{ that are also } q_B)$	0.063	0.209
score-fusion ($\cdot$)	-	0.070	0.111
score-fusion-scale($\cdot$)	-	0.070	0.111
Addition	$\vec{A} + \vec{B}$	0.068	0.222
Max Pool	$\text{MaxPool}(\vec{A}, \vec{B})$	0.069	0.238
CPT (ours)	$\vec{A} \otimes \vec{B}$	0.080	0.268

Table 1: Deriving set-compositional LSR representations on QUEST with Linear Algebra Operations (LAO) in a (zero-shot) setting.

method	representation term scores
<i>Books about non-European monarchs</i>	
vanilla	(monarch##: 1.11, not: 1.1 , book##: 1.03, european: 1.02 , king: 0.38, non: 0.35, ...)
Negation	(monarch##: 1.25, history: 1.13, book##: 1.03, king: 0.77, historical: 0.72, royal: 0.63, ... european: 0.0)
ours	(monarch##: 1.25, history: 1.13, book##: 1.02, king: 0.77, ..., europe: -0.66 , european: -1.17)
<i>(Documentary films about education) $\cap$ (Documentary films about intellectual disability)</i>	
vanilla	(intellectual: 1.14, disability: 1.07, documentary: 1.04, education: 1.04, film##: 0.9, disabled: 0.82, educational: 0.73, ...)
CPT	(education $\cap$ intellectual: 1.57, documentary $\cap$ intellectual: 1.49, education $\cap$ disability: 1.47, documentary $\cap$ disability: 1.40, ...)
finetuned (RQ2)	documentary: 1.21, education: 1.12, disability: 1.08, intellectual: 1.02, disabled: 1.02, school: 0.77, academic: 0.73, autism: 0.67, ...)

Table 2: Examples of set-compositional LSR representations.

docs) and (b) the synthetic nature of Quest queries (minimal overlap between positive and negative query, discussed in Subsection 5.1.1). Orthogonal negation performs on par with Negation Ignore, as it subtracts from $\vec{A}$ the projection of $\vec{B}$ in $\vec{A}$. This does not change the direction of vector $\vec{A}$, and in practice only slightly affects its' norm $|\vec{A}|$, leading to similar results. Table 2 shows example representations demonstrating that in practice, term scores between the two change only slightly. NRF and Disentangled Negation are the most successful methods, with Disentangled Negation having a higher *Recall* compared to NRF. Our hypothesis is that the advantage of these methods in top ranking positions (ie. *NDCG*) stems from their ability to penalize the appearance of negated terms in documents. As shown in Table 2, Disentangled Negation improves *R@100* by specifically penalizing negated terms (e.g., Europe) without reducing the importance of positive ones (e.g., monarch, history). We investigate this property further in Section 5.1.1.

Union ($A \cup B$). Similarly to previous work, we find union queries to be less challenging [31]. This is intuitive, since by nature Union operations are the most easy to satisfy, ie. by relevance to any of the atomic queries. Max Pooling performs slightly better compared to the vanilla variant, with simple Addition being somewhat less effective. This is not surprising, given that Splade models already apply a max pooling operation to aggregate across input tokens.

Intersection ($A \cap B$). Lastly, we find that Intersection is the most challenging operation for LSR models. As explained in Section 3.2.2, traditional lexical methods are unable to capture the combined occurrence of terms across queries that this set operation necessitates. These methods can attribute high scores in all the important terms (eg. education, disability, documentary in Table 2) but fail to capture their combined existence, which would further help them determine relevance. Our Combined Pseudo-Term (CPT) method partially addresses this issue by expanding the vocabulary space to consider two term combinations, outperforming all other methods.

5.1.1 Effectiveness of negation methods for varying levels of interference between positive and negative queries. Previous work has highlighted the challenge of applying negations in IR, namely the difficulty of removing negative aspects from a query vector without weakening positive ones [10, 25]. Widdows [48] showed that as the similarity between positive and negative query vectors increases, it becomes harder to account for negations in ranking. We explore this further in QUEST, which includes queries with varying levels of interference (e.g., "Books about French monarchs but not Napoleon" vs. "1988 fiction books that are not about the environment").

We measure method performance in Figure 3, for varying levels of interference (ie. similarity⁴) between positive and negative query vectors. In low similarity regions, all methods perform relatively on par. However, as the overlap increases, negation methods outperform the *vanilla* variant by a large margin. At the highest level of interference, ignoring the negation or subtracting a constant value of the negative query still work reasonably well, but Disentangled and Orthogonal Negation outperform them by a clear margin. This is due to the ability of these methods to remove the negative aspects of the query without harming the positive aspects, which is a desirable and important property for negations.

⁴Measured with a Dense Retriever (sentence-transformers/all-MiniLM-L6-v2)

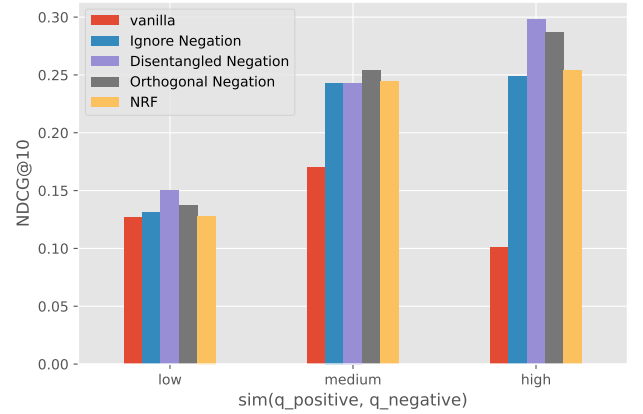


Figure 3: Zero-shot negation method performance, at different interference levels between positive and negative query.

5.1.2 What's the role of expansion and term weighting in the success of linear algebra operations? LSR improves ranking over lexical methods through (a) lexical expansion to bridge query–document vocabulary gaps and (b) term scoring to weight tokens by importance rather than global statistics. We examine how these mechanisms influence LAO's effectiveness.

In Table 3 we incrementally remove (a) query expansion, (b) query term scoring, and (c) document expansion and term scoring from LSR models⁸. By incrementally removing these components from the vanilla variant, we observe a slight performance drop, mirroring the trend in atomic queries. This differs for Set Difference, where a performance drops largely when removing Query Term Scoring (in practice using a binary query vector). This suggests that Splade has (to some extent) learned how to handle negations during their pre-training (MSMarco or BERT pretraining), lowering the scores of negated terms or completely removing them.

Steering our focus towards the improvements of LAO in models with and without expansion and term scoring, we observe that removing Query Expansion does not substantially affect those improvements. However, those improvements collapse when we remove query term scoring (LSR-doc), with the exception of negated queries. For CPT this is expected, as atomic term scores are propagated towards the final query representation through the outer product (Eq. 6), ranking higher documents that contain the CPT (VENEZUELA $\cap$ COLOMBIA), instead of (VENEZUELA $\cap$ ANDES).

We introduced a zero-shot framework for set-compositional representations using LAO. The proposed Disentangled Negation and Combined Pseudo-Term methods outperform the baseline by a wide margin, demonstrating the framework's effectiveness.

5.2 Can retrievers learn query compositional representations?

After investigating zero-shot representations, we explore whether effective representations can be learned by Dense and Learned

⁸Here we use splade-v3 [27], which is also available without expansion & term-scoring

	LSR (full) ⁵	LSR (w/o Q.Exp.) ⁶	LSR-doc ⁷
Query Exp.	✓	✗	✗
Query Term Scoring	✓	✓	✗
Doc. Exp.	✓	✓	✓
Doc. Term Scoring	✓	✓	✓
Query type +Applied Operation	<i>nDCG@10</i> ($\pm \Delta$ <i>nDCG%</i>)		
Atomic	0.284	0.264	0.258
Set Difference	0.155	0.125	0.073
+Disentangled Negation	0.258 (+66.5%)	0.206 (+64.8%)	0.210 (+187%)
Union	0.250	0.236	0.231
+MaxPool	0.262 (+4.8%)	0.251 (+6.4%)	0.233 (+0.9%)
Intersection	0.072	0.065	0.069
+CPT	0.098 (+36.1%)	0.087 (+33.8%)	0.065 (-5.8%)

Table 3: Investigating the effect of (a) Query Expansion, (b) Query Term Scoring, (c) Document Expansion and Document Term Scoring in the effectiveness of LAO. Absolute numbers (first row of group, aligned left) indicate vanilla *nDCG@10* performance for each model, while $\pm \Delta\%$ (aligned right) indicate relative differences after applying the linear algebra operations.

train_set	Atomic		Set Difference		Union		Intersection	
	<i>nDCG@3</i>	<i>R@100</i>	<i>nDCG@3</i>	<i>R@100</i>	<i>nDCG@3</i>	<i>R@100</i>	<i>nDCG@3</i>	<i>R@100</i>
Lexical Retrieval								
–	0.216	0.456	0.052	0.146	0.198	0.368	0.047	0.200
Learned Sparse Retrieval								
Quest-NC ^a	0.219	0.458	0.132	0.307	0.243	0.383	0.063	0.209
Quest-FULL ^b	0.243^a	0.499^a	0.240 ^a	0.409^a	0.260	0.404^{a,c}	0.062	0.227
Quest-NC + LAO ^c (best of RQ1)	–	–	0.258^a	0.398 ^a	0.254	0.387	0.080^{a,b}	0.268^{a,b}
Dense Retrieval								
Quest-NC ^d	0.289	0.571	0.154	0.356	0.278	0.446	0.088	0.290
Quest-FULL ^e	0.311^d	0.577	0.251^d	0.450^d	0.291^d	0.466^d	0.081	0.283

Table 4: To what extent can different retriever families learn compositional representations from training? (RQ2). Superscripts indicate significant improvements, compared to retrievers of the same family (p-value<0.05, paired t-test). (QUEST dataset)

Sparse Retrievers from training. To answer **RQ2**, we finetune retrievers on (a) Quest-NC, the non-compositional subset of Quest containing atomic queries and (b) Quest-FULL, that also contains set-compositional queries and present results in Table 4. In general, we observe that Dense Retrievers seem to have a better performance in Quest, both when it comes to atomic or compositional queries.

Retrievers can effectively learn to handle negated queries, but not intersections. Adding set-compositional queries to the training set seems to improve performance on atomic queries by roughly 10%, likely due to a larger training set. Large performance gains are observed only for set difference queries, where LSR learns to remove negated terms from query representations (Table 2). Improvements for union are significant, but comparable to the ones observed for atomic. Interestingly, none of the retrievers seem to learn from training on intersections. This aligns with prior research showing that Dense Retrievers struggle with intersections [7, 31], but further reveals that, under some domain shift (same dataset, different entity

type), both Dense and LSR models fail to effectively model intersections. This is expected for LSR, given their limitations in forming term combinations (Section 3.2.2), but surprising for DR models, which generate high-dimensional representations without sparsity or lexical constraints. To further solidify these findings, we also experimented with training using hard negatives (specifically BM25 and compositional negatives), but found no substantial differences.

Zero-shot vs. finetuned representations Finally, we observe that applying LAO between LSR embeddings achieves performance comparable or better than training them for compositionality, indicating that their representation power may be in some ways limited. For intersections, solving this representation bottleneck would require expanding the vocabulary (ie. Combined Pseudo Terms), or abandoning the semantic relationship between sparse dimensions and tokens leading to a Dense Retriever with more dimensions. Similarly, we observe that the zero-shot *Disentangled Negation* outperforms both trained LSR and DR models on Set Difference queries in *nDCG* and remains competitive in Recall. Our hypothesis is that

Model	FT on NevIR	Negative term scores	+/- weight aggreg.	Pairwise Accuracy
random	✗	–	–	25.00%
contriever	✗	–	–	9.62%
	✓	–	–	42.88%
splade	✗	✗	–	7.88%
	✓	✗	–	23.07%
	✓	✓	$\pm max $	37.09%
	✓	✓	<i>sum</i>	42.73%

Table 5: Performance of LSR models trained with negative term scores on NevIR.

Nega- tive scores	+/- weight aggreg.	Atomic		Set Difference	
		nDCG@10	R@100	nDCG@10	R@100
✗	–	0.284	0.540	0.205	0.413
✓	$\pm max $	0.182	0.418	0.210	0.340
✓	<i>sum</i>	0.165	0.391	0.218	0.351

Table 6: Performance of LSR models trained with negative term scores on Quest.

negative term weights benefit ranking, as they allow the scoring function to go beyond ignoring negated terms (eg. “european”, Table 2) and penalize documents that contain them.

5.3 Improving LSR representations for negations

Previous experiments indicated that LSR representations that assign *negative term scores* to negated aspects of a query (ie. Disentangled Negation, NRF) show an advantage over methods that ignore negations. LSR models seem well suited for addressing the problem of interference between positive and negative queries. Therefore, we explore **RQ3**, aimed at improving the representation power of LSR models by allowing them to attribute *negative term scores*.

Table 5 measures performance in NevIR [47], showing that even after finetuning, LSR models without negative term weights fail to surpass the random baseline (23.07% vs 25%). In contrast, allowing splade models to score terms negatively helps them almost double their performance on NevIR (42.73% vs. 23.07%). Aggregating the maximum positive and negative query scores by adding them (*sum* weight aggregation), leads to better performance compared to selecting the absolute maximum ($\pm|max|$ weight aggregation). This is possibly due to the fact that, in this variant, gradients flow back from both the positive and negative part of the query to the *out* embedding, leading to a more informative training procedure.

Table 6 reports ranking performance on the Quest dataset. We observe that models with negative weights struggle on atomic queries and do not perform on par with the baseline that uses only positive weights. On Set Difference queries, allowing negative term scores improves performance for *nDCG* but degrades in Recall. We hypothesize that the observed degradation arises because changing the Splade activation function (Figure 2, Equations 2, 12, and 13) introduces a drastic architectural change that is difficult to learn on QUESTs’ relatively small training set (236 negative queries).

That is due to pre-training of LSR models in the Masked Language Modelling task (BERT [8]), as well as their pre-finetuning in ranking, where models are trained to predict very low values on the MLM output out_{ij} (left region of Figure 2) for irrelevant terms, and large positive values (right region, Figure 2) for both positive and negative terms (see “non-European” for *vanilla* variant, Table 2). Therefore, when model weights are initialized using the SNReLU activation function, it becomes necessary to re-calibrate the learned notions of *relevance*, *negative relevance*, and *irrelevance*. This requires shifting negatively relevant terms from the positive region of Figure 2 to the negative (left region), and irrelevant terms, that hold large negative values to the region around 0 ($out_{ij} \in [-\epsilon, +\epsilon]$) so that the representation remains sparse. It appears that such an adaptation is possible in NevIR, where a relatively larger training set with hard contrastive negatives exists, but is challenging to achieve in QUEST.

To conclude, we find that the representation power of LSR models can be improved by allowing models penalize negatively relevant terms, given a large enough dataset to finetune model weights.

6 Conclusions and future directions

In this work, we focused on constructing set-compositional and negated representations for first stage ranking, primarily focusing on LSR representations that are lexically grounded and easier to manipulate. We found that constructing effective representations is possible using a zero-shot framework that combines atomic representations using Linear Algebra Operations. We introduced LAO that target specific set operations, more importantly: *Disentangled Negation* for Set-Difference, that aims to tackle the challenging problem of interference between positive and negative queries, and Combined Pseudo-Terms that aims to expand the representation space of Intersections by considering term combinations. We found both to be particularly effective, performing on par or better compared to models trained for set-compositionality. When it comes to training, we found that both LSR and DR cannot learn effective representations for intersections, but do benefit from training on queries with negations. Finally, we adapted LSR models for attributing negative term scores to negated aspects, that in practice allows them to effectively penalize documents containing those aspects. This proved particularly effective when a reasonably sized negation dataset is available, but more challenging when data is scarce.

This work lays the foundation for creating lexically grounded, set-compositional and negated LSR representations. We identify several exciting future directions, including: (a) researching ways on making LSR models with negative term scores more effective under data scarce settings, possibly by pre-training methods and data generation using LLMs, or knowledge distillation from existing retrievers and rerankers, (b) exploring the efficacy of late-interaction models in creating set-compositional representations, due to their more expressive representations (ie. one Dense vector per query token), (c) investigating further the problem of query decomposition with LLMs and exploring whether a more efficient query encoder can also perform this task with high accuracy.

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GenAI disclosure statement

Generative AI tools (ChatGPT) were used for minor editing, such as rephrasing and compression, while writing this paper. The same GenAI tools were also employed in preliminary experiments, showing that LLMs can decompose complex queries into atomic components in QUEST with high accuracy.

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